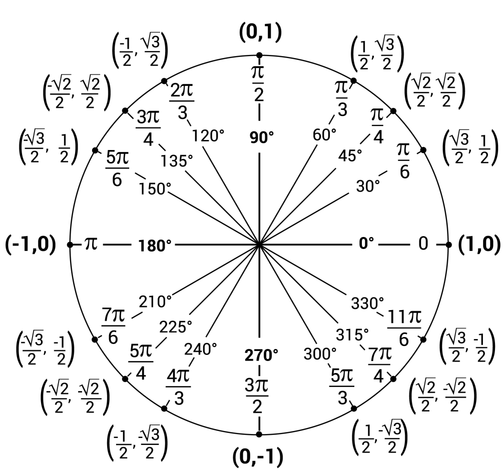
1 Useful Identities

Complex numbers and trigonometry

- Complex $z = x + jy = |z|e^{j\angle z}$. $|z| = \sqrt{x^2 + y^2}$, $\angle z = \operatorname{atan2}(y, x)$.
- Magnitude/Angle Properties: If $z = z_1 z_2 / z_3$, then $|z| = |z_1||z_2|/|z_3|$ and $\angle z = \angle z_1 + \angle z_2 - \angle z_3$.
- Triangle Inequality: For complex numbers z_1, z_2:
  - $|z_1 + z_2| \le |z_1| + |z_2|$
  - $||z_1| - |z_2|| \le |z_1 - z_2|$ (Reverse Triangle Inequality)
- Euler's formula: $e^{j\theta} = \cos\theta + j\sin\theta$
- Common Phasor Identities: $e^{j0} = 1$; $e^{j\pi/2} = j$; $e^{j\pi} = -1$; $e^{-j\pi/2} = -j$. Also: $e^{j\pi/4} = \frac{1+j}{\sqrt{2}}$; $e^{j\pi/6} = \frac{\sqrt{3}}{2} + j\frac{1}{2}$; $e^{j\pi/3} = \frac{1}{2} + j\frac{\sqrt{3}}{2}$. $e^{j(\pi+2k\pi)} = -1$; $e^{j(2k\pi)} = 1$
- Nth roots of a complex number: If $z^N = w = r_0 e^{j\theta_0}$, then $z_k = r_0^{1/N} e^{j(\theta_0 + 2\pi k)/N}$ for $k = 0, 1, \dots, N-1$. (Roots are equally spaced on a circle).
- Basic trigonometric identities:
  $\cos\theta = \frac{1}{2}(e^{j\theta} + e^{-j\theta})$; $\sin\theta = \frac{1}{2j}(e^{j\theta} - e^{-j\theta})$
  $\operatorname{sinc}(t) = \frac{\sin(\pi t)}{\pi t}$
  $\sin^2(\theta) + \cos^2(\theta) = 1$
  $\sin(\theta) = \cos(\theta - \pi/2)$
  $\cos(k\pi) = (-1)^k$ for integer k.
  $e^{j(A+B)} = \cos(A+B) + j\sin(A+B)$
  $e^{jA}e^{jB} = (\cos(A) + j\sin(A))(\cos(B) + j\sin(B))$
  $= (\cos(A)\cos(B) - \sin(A)\sin(B)) + j(\sin(A)\cos(B) + \cos(A)\sin(B))$
  $\cos(A+B) = \cos(A)\cos(B) - \sin(A)\sin(B)$
  $\sin(A+B) = \sin(A)\cos(B) + \cos(A)\sin(B)$
  $\cos(A-B) = \cos(A)\cos(B) + \sin(A)\sin(B)$
  $\sin(A-B) = \sin(A)\cos(B) - \cos(A)\sin(B)$
  $\sin(A)\cos(B) = \frac{1}{2}(\sin(A+B) + \sin(A-B))$
  $\cos(A)\sin(B) = \frac{1}{2}(\sin(A+B) - \sin(A-B))$
  $\cos(A)\cos(B) = \frac{1}{2}(\cos(A+B) + \cos(A-B))$
  $\sin(A)\sin(B) = \frac{1}{2}(\cos(A-B) - \cos(A+B))$
  $\angle\cos(\omega) = \begin{cases} \pi & \text{if } -\pi \le \omega < -\frac{\pi}{2} \\ 0 & \text{if } -\frac{\pi}{2} < \omega < \frac{\pi}{2} \\ \pi & \text{if } \frac{\pi}{2} < \omega \le \pi \end{cases}$
  $\angle\sin(\omega) = \begin{cases} 0 & \text{if } 0 < \omega < \pi \\ \pi & \text{if } -\pi < \omega < 0 \end{cases}$
- Infinite geometric series: $\sum_{n=0}^{\infty} z^n = \frac{1}{1-z}$, for $|z| < 1$; $\sum_{n=0}^{\infty} nz^n = \frac{z}{(1-z)^2}$, for $|z| < 1$.
- Finite geometric series:
  $\sum_{n=0}^{N-1} z^n = \frac{1-z^N}{1-z}$ for $z \ne 1$. (Sum is N if $z = 1$).
  $\sum_{k=0}^{N} z^k = \frac{1-z^{N+1}}{1-z}$ for $z \ne 1$. (Sum is $N+1$ if $z = 1$).
- Useful identities involving complex exponentials:
  $1 + e^{-jM} = e^{-j\frac{M}{2}}(e^{j\frac{M}{2}} + e^{-j\frac{M}{2}}) = 2e^{-j\frac{M}{2}}\cos\left(\frac{M}{2}\right)$
  $1 - e^{-jM} = e^{-j\frac{M}{2}}(e^{j\frac{M}{2}} - e^{-j\frac{M}{2}}) = 2je^{-j\frac{M}{2}}\sin\left(\frac{M}{2}\right)$



2 Signals - Chapter 1

- Signal energy and power
  Total energy (CT): $E_\infty = \int_{-\infty}^{\infty} |x(t)|^2 dt$
  Total energy (DT): $E_\infty = \sum_{n=-\infty}^{\infty} |x[n]|^2$
  Average power (CT): $P_\infty = \lim_{T\to\infty} \frac{1}{2T}\int_{-T}^{T} |x(t)|^2 dt$
  Average power (DT): $P_\infty = \lim_{N\to\infty} \frac{1}{2N+1}\sum_{n=-N}^{N} |x[n]|^2$
  For periodic signals:
  Average power in one period (CT): $\frac{1}{T}\int_T |x(t)|^2 dt$
  Average power in one period (DT): $\frac{1}{N}\sum_{n=\langle N\rangle} |x[n]|^2$
- Transformations of independent variable
  - Time shifting, time reversal, time scaling
  - Interpretations of $x(at - b) = x(a(t - b/a))$:
    * Shift by b then scale t by a: $x(t-b) \to x(at-b)$.
    * Scale t by a then shift t by b/a: $x(at) \to x(a(t - b/a))$.
  - Interpretations of $x[an - b]$, a, b integers (similar logic)
- Periodic signals
  - Periodicity conditions:
    * CT: $x(t)$ is periodic if $\exists T > 0$ s.t. $x(t+T) = x(t)$ for all t. Smallest such T is the fundamental period T_0.
    * DT: $x[n]$ is periodic if $\exists N \in \mathbb{Z}, N > 0$ s.t. $x[n+N] = x[n]$ for all n. Smallest such N is the fundamental period N_0.
  - Fundamental frequency $\omega_0 = 2\pi/T_0$ (CT), $\Omega_0 = 2\pi/N_0$ (DT).
  - Finding Fund. Period (T_0 or N_0):
    * Single CT signal (e.g., $e^{j\omega_0 t}, \cos(\omega_0 t)$): $T_0 = 2\pi/|\omega_0|$.
    * Single DT signal $x[n] = e^{j\Omega_0 n}$: Ω_0 must be a rational multiple of 2π, i.e., $\Omega_0 = 2\pi\frac{m}{N}$ ($m, N \in \mathbb{Z}$, $N > 0$, m, N coprime). Fund. period $N_0 = N$. If $\Omega_0 = 0$ (DC signal), $N_0 = 1$.
    * Sum of CT signals $x(t) = \sum_i x_i(t)$ with fund. periods T_i: $T_0 = \operatorname{LCM}(T_1, T_2, \dots)$ if all T_i/T_j are rational. If $x(t) = \sum_i C_i e^{j\omega_i t}$ and all ω_i/ω_j are rational, let $\omega_f = \gcd(|\omega_1|, |\omega_2|, \dots)$, then $T_0 = 2\pi/\omega_f$. Otherwise, not periodic.
    * Sum of DT signals $x[n] = \sum_i x_i[n]$ with fund. periods N_i: $N_0 = \operatorname{LCM}(N_1, N_2, \dots)$.
  - Periodicity and scaling
- Even and Odd signals
  - Definitions of even and odd signals:
    * Even signal: $x(t) = x(-t)$ (CT), $x[n] = x[-n]$ (DT).
    * Odd signal: $x(t) = -x(-t)$ (CT), $x[n] = -x[-n]$ (DT).
  - Even-odd decomposition theorem: $x(t) = \mathcal{E}v\{x(t)\} + \mathcal{O}d\{x(t)\}$ where $\mathcal{E}v\{x(t)\} = \frac{x(t)+x(-t)}{2}$ and $\mathcal{O}d\{x(t)\} = \frac{x(t)-x(-t)}{2}$. (Similar for DT).
- CT and DT impulse and unit step signals
  Relationships
  $\sum_{n=-\infty}^{\infty} \delta[n] = 1$
  $\delta[n] = u[n] - u[n-1]$
  $u[n] = \sum_{k=0}^{\infty} \delta[n-k] = \sum_{k=-\infty}^{n} \delta[k]$
  $\int_{-\infty}^{\infty} \delta(t)dt = 1$
  $\delta(t) = \frac{d}{dt}u(t)$
  $u(t) = \int_{-\infty}^{t} \delta(\tau)d\tau$
  Sampling and Sifting Properties
  - Sampling Property:
    * $x[n]\delta[n-n_0] = x[n_0]\delta[n-n_0]$
    * $x(t)\delta(t-t_0) = x(t_0)\delta(t-t_0)$
  - Sifting Property:
    * $\sum_{n=-\infty}^{\infty} x[n]\delta[n-n_0] = x[n_0]$
    * $\int_{-\infty}^{\infty} x(t)\delta(t-t_0)dt = x(t_0)$

  Representation property
  $x[n] = \sum_{k=-\infty}^{\infty} x[k]\delta[n-k]$
  $x(t) = \int_{-\infty}^{\infty} x(\tau)\delta(t-\tau)d\tau$
- Complex exponential signals
  CT: $x(t) = ce^{at}, c, a \in \mathbb{C}$
  DT: $x[n] = ca^n, c, a \in \mathbb{C}$
  $x(t) = e^{j\omega_0 t}$ periodic, fund. frequency ω_0, fund. period $T = 2\pi/|\omega_0|$
  $x[n] = e^{j\Omega_0 n}$ periodic in n if and only if $\Omega_0 = 2\pi m/N$, for $m, N \in \mathbb{Z}, N > 0$. If $\gcd(|m|, N) = 1$, fund. period is N.
  $x[n] = e^{j\Omega n}$ periodic in Ω, period 2π.

3 Systems - Chapters 1 and 2

- Basic system properties
  Memoryless: $y(t_0)$ depends only on $x(t_0)$ (or $y[n_0]$ on $x[n_0]$).
  Invertible: Distinct inputs produce distinct outputs.
  Causal: $y(t_0)$ depends only on $x(t)$ for $t \le t_0$ (or $y[n_0]$ on $x[k]$ for

$k \le n_0$).
Stable (BIBO): If $|x(t)| \le \alpha < \infty \Rightarrow |y(t)| \le \beta < \infty$.
Time-invariant: Input $x(t-t_0) \Rightarrow$ output $y(t-t_0)$.
Linear: $S(ax_1 + bx_2) = aS(x_1) + bS(x_2)$.

- System impulse response $h(t)$ and step response $s(t)$.
- LTI systems
  The impulse response $h(t)$ or $h[n]$ characterizes the system.
  CT: $y(t) = x(t) * h(t)$ DT: $y[n] = x[n] * h[n]$
- Relationship between impulse response and step response
  $s(t) = \int_{-\infty}^{t} h(\tau)d\tau \quad h(t) = \frac{ds(t)}{dt}$
  $s[n] = \sum_{k=-\infty}^{n} h[k] \quad h[n] = s[n] - s[n-1]$
- Convolution formulas (DT and CT)
  $y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k] \quad y(t) = \int_{-\infty}^{\infty} x(\tau)h(t-\tau)d\tau$
- Properties of convolution
  Commutativity, associativity, distributivity over addition
  $x(t) * \delta(t-t_0) = x(t-t_0); \quad x[n] * \delta[n-n_0] = x[n-n_0]$
  $x(t) * u(t) = \int_{-\infty}^{t} x(\tau)d\tau; \quad x[n] * u[n] = \sum_{k=-\infty}^{n} x[k]$
- Convolution of Causal Exponential Sequences
  $a^n u[n] * b^n u[n] = \frac{a^{n+1} - b^{n+1}}{a-b} u[n]$, for $a \neq b$.
- Impulse response of serial and parallel LTI systems.
  Serial: $h(t) = h_1(t) * h_2(t)$ Parallel: $h(t) = h_1(t) + h_2(t)$
- Impulse response and properties of LTI systems
  Memoryless: $h(t) = a\delta(t)$ (CT); $h[n] = a\delta[n]$ (DT)
  Invertible: $\exists g(t)$ s.t. $h(t) * g(t) = \delta(t)$.
  Causal: $h(t) = 0$ for $t < 0$; $h[n] = 0$ for $n < 0$.
  Stable (BIBO): $\int_{-\infty}^{\infty} |h(t)|dt < \infty$; $\sum_{n=-\infty}^{\infty} |h[n]| < \infty$.
- Differentiation property of CT LTI systems: $x(t) \to y(t) \Rightarrow \frac{dx(t)}{dt} \to \frac{dy(t)}{dt}$
- LTI systems defined by differential/difference equations
  $\sum_{k=0}^{N} a_k \frac{d^k y(t)}{dt^k} = \sum_{k=0}^{M} b_k \frac{d^k x(t)}{dt^k}$
  $\sum_{k=0}^{N} a_k y[n-k] = \sum_{k=0}^{M} b_k x[n-k]$
- Filters
  - Non-recursive filters (FIR): $y[n] = \sum_{k=0}^{M} b_k x[n-k]$.
  - DT blur filter: $h[n] = \frac{1}{N}\sum_{k=0}^{N-1} \delta[n-k] = \frac{1}{N}(u[n] - u[n-N])$.
  - Recursive filters (IIR): $\sum a_k y[n-k] = \sum b_k x[n-k]$.
  - DT first order filter: $y[n] - ay[n-1] = x[n]$, $|a| < 1$. Impulse response $h[n] = a^n u[n]$. If $a > 0$, low-pass. If $a < 0$, high-pass (alternating polarity).

4 CT and DT Fourier Series - Chapter 3

- Key equations
  CTFS for periodic $x(t)$, period T, $\omega_0 = 2\pi/T$
  Synthesis: $x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$
  Analysis: $a_k = \frac{1}{T}\int_T x(t)e^{-jk\omega_0 t}dt$
  DTFS for periodic $x[n]$, period N, $\Omega_0 = 2\pi/N$
  Synthesis: $x[n] = \sum_{k=\langle N \rangle} a_k e^{jk\Omega_0 n}$
  Analysis: $a_k = \frac{1}{N}\sum_{n=\langle N \rangle} x[n]e^{-jk\Omega_0 n}$
- Response of LTI system to complex exponential
  - Eigenfunctions: e^{st} (CT), z^n (DT).
  - CT: $e^{st} \to H(s)e^{st}$, where $H(s) = \int_{-\infty}^{\infty} h(t)e^{-st}dt$ (Laplace Transform of $h(t)$). Convergence determined by ROC.
  - DT: $z^n \to H(z)z^n$, where $H(z) = \sum_{k=-\infty}^{\infty} h[k]z^{-k}$ (Z-Transform of $h[k]$). Convergence determined by ROC.
- How to determine if system may be LTI by action on complex exponential signals: Check if eigenfunction property is satisfied. If violated, system is not LTI.
- Filtering of periodic signal through an LTI system
  $x(t) = \sum a_k e^{jk\omega_0 t} \to y(t) = \sum a_k H(jk\omega_0) e^{jk\omega_0 t}$
  $x[n] = \sum a_k e^{jk\Omega_0 n} \to y[n] = \sum a_k H(e^{jk\Omega_0}) e^{jk\Omega_0 n}$
- Properties of CTFS/DTFS (Tables 3.1 & 3.2)
  Linearity, Time/Freq Shifting, Time Reversal/Scaling, Convolution, Multiplication, Parseval's Relation, Differentiation/Integration (CT), First Difference/Running Sum (DT).
- FS and signal properties: x real $\Leftrightarrow a_k = a^*_{-k}$; x real & even $\Leftrightarrow a_k$ real & even; x real & odd $\Leftrightarrow a_k$ imaginary & odd.
- Finding system function from differential/difference equations
  $H(s) = \frac{\sum_{k=0}^{M} b_k s^k}{\sum_{k=0}^{N} a_k s^k}$, then $s = j\omega$ for $H(j\omega)$.
  $H(z) = \frac{\sum_{k=0}^{M} b_k z^{-k}}{\sum_{k=0}^{N} a_k z^{-k}}$, then $z = e^{j\omega}$ for $H(e^{j\omega})$.
- Effect of LTI system with real h on sinusoids
  $\cos(\omega_0 t) \to |H(j\omega_0)| \cos(\omega_0 t + \angle H(j\omega_0))$

5 CT Fourier Transform - Chapter 4

- Key equations
  - Synthesis: $x(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)e^{j\omega t}d\omega$
    (e.g., to find $x(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} X(j\omega)d\omega$)
  - Analysis: $X(j\omega) = \int_{-\infty}^{\infty} x(t)e^{-j\omega t}dt$
    (e.g., to find DC content $X(j0) = \int_{-\infty}^{\infty} x(t)dt$)
- Properties of CTFT (Table 4.1): Linearity, Time/Freq Shifting, Conjugation, Time Reversal, Time Scaling, Convolution, Multiplication, Differentiation, Integration.
- Parseval's relation: $\int_{-\infty}^{\infty} |x(t)|^2 dt = \frac{1}{2\pi}\int_{-\infty}^{\infty} |X(j\omega)|^2 d\omega$
- FT and signal properties:
  - $x(t)$ real $\Leftrightarrow X(j\omega) = X^*(-j\omega)$ (conjugate symmetric)
  - $x(t)$ real and even $\Leftrightarrow X(j\omega)$ real and even
  - $x(t)$ real and odd $\Leftrightarrow X(j\omega)$ purely imaginary and odd
  - For real signals: $\mathcal{E}v\{x(t)\} \leftrightarrow \mathrm{Re}\{X(j\omega)\}$, $\mathcal{O}d\{x(t)\} \leftrightarrow j\mathrm{Im}\{X(j\omega)\}$
- Basic CTFT Pairs (Table 4.2)
  - Periodic $x(t) = \sum a_k e^{jk\omega_0 t} \leftrightarrow X(j\omega) = 2\pi \sum a_k \delta(\omega - k\omega_0)$
  - $x(t) = 1 \leftrightarrow 2\pi\delta(\omega)$
  - Periodic impulse train (picket fence) $\leftrightarrow$ periodic impulse train
  - Rectangular pulse $\leftrightarrow$ sinc function
  - Sinc function $\leftrightarrow$ rectangular pulse
  - $e^{-at}u(t), \mathrm{Re}\{a\} > 0 \leftrightarrow \frac{1}{a+j\omega}$
- Systems from diff. eqns: $H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{\sum b_k (j\omega)^k}{\sum a_k (j\omega)^k}$
- Filtering: $y(t) = x(t) * h(t) \leftrightarrow Y(j\omega) = X(j\omega)H(j\omega)$
- Inverse systems: $h(t) * g(t) = \delta(t) \leftrightarrow H(j\omega)G(j\omega) = 1$

6 DT Fourier Transform - Chapter 5

- Key equations
  - Synthesis: $x[n] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})e^{j\omega n}d\omega$
    (e.g., $x[0] = \frac{1}{2\pi}\int_{2\pi} X(e^{j\omega})d\omega$)
  - Analysis: $X(e^{j\omega}) = \sum_{n=-\infty}^{\infty} x[n]e^{-j\omega n}$
    (e.g., $X(e^{j0}) = \sum_{n=-\infty}^{\infty} x[n]$)
- Periodicity: $X(e^{j\omega})$ is always periodic in ω with period 2π.
- Properties of DTFT (Table 5.1): Similar to CTFT.
- Basic DTFT Pairs (Table 5.2):
  - Periodic $x[n] = \sum a_k e^{jk\omega_0 n} \leftrightarrow X(e^{j\omega}) = 2\pi \sum_{l=-\infty}^{\infty} \sum_{k=\langle N \rangle} a_k \delta(\omega - k\omega_0 - 2\pi l)$
  - Others similar to CTFT (e.g., $a^n u[n] \leftrightarrow \frac{1}{1-ae^{-j\omega}}$ for $|a| < 1$)
- DTFT of a Shifted Causal Exponential
  $a^n u[n-n_0] \overset{\text{DTFT}}{\longleftrightarrow} \frac{a^{n_0} e^{-j\omega n_0}}{1-ae^{-j\omega}}$
- Systems from diff. eqns: $H(e^{j\omega}) = \frac{\sum b_k e^{-j\omega k}}{\sum a_k e^{-j\omega k}}$. Inverse systems swap roles of $x[n], y[n]$. Use PFE and tables to find $h[n]$.
- Filtering: $y[n] = x[n] * h[n] \leftrightarrow Y(e^{j\omega}) = X(e^{j\omega})H(e^{j\omega})$
- Filter frequency response derivation:
  - Length-N blur filter: $H(e^{j\omega}) = \frac{1}{N}e^{-j\omega(N-1)/2}\frac{\sin(\omega N/2)}{\sin(\omega/2)}$
  - Recursive filter $y[n] - ay[n-1] = x[n]$: $H(e^{j\omega}) = \frac{1}{1-ae^{-j\omega}}$. $a > 0$: low-pass, $a < 0$: high-pass.

7 Amplitude Modulation - Chapter 8

- Modulation: For a signal $x(t)$ bandlimited to $[-W, W]$ and carrier frequency $\omega_c > W$:
  - With cosine carrier $y(t) = x(t)\cos(\omega_c t)$, the transform is $Y(j\omega) = \frac{1}{2}[X(j(\omega - \omega_c)) + X(j(\omega + \omega_c))]$.
  - With sine carrier $y(t) = x(t)\sin(\omega_c t)$, the transform is $Y(j\omega) = \frac{1}{2j}[X(j(\omega - \omega_c)) - X(j(\omega + \omega_c))]$.
- Demodulation: $z(t) = y(t)c(t) = x(t)\cos^2(\omega_c t)$
  - $Z(j\omega) = \frac{1}{2}Y(j\omega) * C(j\omega) = \frac{1}{4}X(j(\omega - 2\omega_c)) + \frac{1}{2}X(j\omega) + \frac{1}{4}X(j(\omega + 2\omega_c))$
- Signal recovery: If $\omega_c > W$, pass $z(t)$ through an ideal LPF with cutoff $\omega_{co} \in (W, 2\omega_c - W)$ and gain 2.
- Effects of demodulating with different sinusoid:
  - Freq mismatch: $\cos(\omega_c t)\sin(\omega_e t)$ with $\omega_c \neq \omega_e$.
  - Phase mismatch: $\cos(\omega_c t)\sin(\omega_c t) = \frac{1}{2}\sin(2\omega_c t)$. LPF output is 0.

8 Sampling Theory - Chapter 7

- **Impulse train sampling**
  CT signal, band-limited to $[-W, W]$: $x(t) \leftrightarrow X(j\omega), X(j\omega) = 0$, for $|\omega| > W$
  Impulse train, sampling period T, sampling frequency $\omega_s = 2\pi/T$: $p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT) \leftrightarrow P(j\omega) = \frac{2\pi}{T} \sum_{k=-\infty}^{\infty} \delta(\omega - k\omega_s)$
  Sampled signal: $x_p(t) = x(t)p(t) = \sum_{n=-\infty}^{\infty} x(nT)\delta(t - nT) \leftrightarrow X_p(j\omega) = \frac{1}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - k\omega_s))$
- **Sampling Theorem:**
  Let $x(t)$ be band-limited to $[-W, W]$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n \in \mathbb{Z}$ if $\omega_s = 2\pi/T > 2W$.
  The signal $x(t)$ can be reconstructed from the signal $x_p(t)$ using a **low-pass** filter with cutoff frequency $\omega_0 = \omega_s/2 = \pi/T$ and gain T.
  If $\omega_s < 2W$, then **aliasing** occurs, and $x(t)$ cannot be reconstructed from $x_p(t)$.
- Using CTFT properties and transform pairs to determine the maximum frequency of a signal $y(t)$ obtained from $x(t)$ by various operations, e.g., conjugation, time-shifting, combining with another signal by linear operation, conjugation, multiplication, etc.
- **DT processing of CT signals**
  Define $x(t), p(t), x_p(t)$ as above.
  Define $x_d[n] = x(nT), n \in \mathbb{Z}$
  Using formula for $x_p(t)$, rewrite $X_p(j\omega) = \sum_{n=-\infty}^{\infty} x(nT)e^{-j\omega nT}$
  Using DTFT, $X_d(e^{j\Omega}) = \sum_{n=-\infty}^{\infty} x_d[n]e^{-j\Omega n} = \sum_{n=-\infty}^{\infty} x(nT)e^{-j\Omega n}$
  Thus, $X_d(e^{j\Omega}) = X_p(j\Omega/T)$ or $X_d(e^{j\omega T}) = X_p(j\omega)$.
  The plot of $X_d(e^{j\Omega})$ is obtained from the plot of $X_p(j\omega)$ by multiplying the ω-axis labels of $X_p(j\omega)$ by T. So, $\omega_s = 2\pi/T$ becomes $\Omega = 2\pi$.
- **Example of aliasing when undersampling a sinusoidal signal**
  Change of frequency and possible phase shift (reversal)
- **Alternating impulse train sampling technique**
  $x_p(t) = \sum_{n=-\infty}^{\infty} x(nT)(-1)^n\delta(t - nT) \leftrightarrow X_p(j\omega) = \frac{2}{T} \sum_{k=-\infty}^{\infty} X(j(\omega - (2k+1)\omega_s))$
- **Bandpass sampling**
  For $X(j\omega) = 0$ outside $\omega_L \le |\omega| \le \omega_H$, sampling frequency $\frac{2\omega_H}{n} \le \omega_s \le \frac{2\omega_L}{n-1}$ for any $1 \le n \le \lfloor \frac{\omega_H}{\omega_H - \omega_L} \rfloor$ allows reconstruction.

9 Laplace Transform - Chapter 9

- Laplace Transform: $X(s) = \int_{-\infty}^{\infty} x(t)e^{-st}dt$, for $s = \sigma + j\omega \in \mathbb{C}$.
- Region of Convergence (ROC): The set of $s \in \mathbb{C}$ for which $X(s)$ exists.
- Relation to FT: If ROC includes the $j\omega$-axis ($\sigma = 0$), then $X(j\omega)$ is the FT of $x(t)$.
- Basic LT examples:
  - Right-sided: $e^{-at}u(t) \leftrightarrow \frac{1}{s+a}$, ROC: $\mathrm{Re}\{s\} > -\mathrm{Re}\{a\}$.
  - Left-sided: $-e^{-at}u(-t) \leftrightarrow \frac{1}{s+a}$, ROC: $\mathrm{Re}\{s\} < -\mathrm{Re}\{a\}$.
- Rational $X(s)$, poles, zeros, and pole-zero plots.
- The 8-fold way of the ROC:
  - ROC is composed of strips parallel to the $j\omega$-axis.
  - ROC does not contain any poles.
  - If $x(t)$ is finite duration, ROC is the entire s-plane.
  - If $x(t)$ is right-sided, ROC is a right half-plane.
  - If $x(t)$ is left-sided, ROC is a left half-plane.
  - If $x(t)$ is two-sided, ROC is a vertical strip.
- ROC and LTI system properties for system $H(s)$:
  - Causal $\Rightarrow$ ROC is a right half-plane. (If rational, RHP to the right of rightmost pole).
  - Stable $\Leftrightarrow$ ROC contains the $j\omega$-axis.
  - Causal and Stable $\Leftrightarrow$ All poles are in the left half-plane.
- Inverse LT: Use partial fraction expansion (PFE) and tables of LT pairs.
- Geometric evaluation of Fourier Transform using pole-zero plot.
  - Magnitude $|H(j\omega)|$ using lengths of vectors from poles and zeros to $j\omega$, i.e., $(j\omega - \rho)$ and $(j\omega - \zeta)$.
  - Phase $\angle H(j\omega)$ using angles of vectors from poles and zeros to $j\omega$, i.e., $(j\omega - \rho)$ and $(j\omega - \zeta)$.
- Properties of LT (Table 9.1): Linearity, Time Shifting, Shifting in s-domain, Time Scaling, Conjugation, Convolution, Differentiation, Integration.
- Analysis of LTI systems from diff. eqns:
  - System function: $H(s) = \frac{Y(s)}{X(s)} = \frac{\sum b_k s^k}{\sum a_k s^k}$.
  - Convert to rational $H(s)$, use PFE, invert using tables. ROC is determined by system properties (causal, stable, etc.).
- System function algebra:
  - Serial: $H(s) = H_1(s)H_2(s)$
  - Parallel: $H(s) = H_1(s) + H_2(s)$
  - Feedback (positive): $H(s) = \frac{H_{ff}(s)}{1 - H_{ff}(s)H_{fb}(s)}$
- **Block Diagram (Direct Form II)** for $H(s) = \frac{N(s)}{D(s)}$: To draw the diagram for $H(s) = \frac{\sum b_k s^k}{\sum a_k s^k}$:
  - Decompose into $H(s) = H_1(s)H_2(s)$ where $H_1(s) = N(s)$ (zeros) and $H_2(s) = 1/D(s)$ (poles).
  - Let $Y(s) = H_1(s)[H_2(s)X(s)]$. Define an intermediate signal $F(s) = H_2(s)X(s)$.
  - **Feedback part** (from $F(s)D(s) = X(s)$): Rearrange for the highest derivative term (assuming $a_N = 1$): $s^N F(s) = X(s) - \sum_{k=0}^{N-1} a_k s^k F(s)$. This defines the integrator chain and feedback loops (scaled by $-a_k$) that create $F(s)$ from input $X(s)$.
  - **Feedforward part** (from $Y(s) = N(s)F(s)$): Expand as $Y(s) = \sum_{k=0}^{M} b_k s^k F(s)$. The output $Y(s)$ is a weighted sum (scaled by b_k) of the signals tapped from the integrator chain.

10 Extra Junk

- Unity gain means gain of 1

11 Examples

2022 Problem 3 [Modulation]

Givens:

- $x(t) = \frac{\sin(20t)}{\pi t} \leftrightarrow X(j\omega) = \begin{cases} 1, & |\omega| < 20 \\ 0, & |\omega| > 20 \end{cases}$
- $y(t) = x(t)\cos(30t)$
- $w(t)$ is the output of $y(t)$ passed through BPF: $H_{BP}(j\omega) = 1$ for $30 < |\omega| < 50$.
- $z(t) = w(t)\cos(40t)$
- $r(t)$ is the output of $z(t)$ passed through LPF with gain 2 and cutoff $\omega_c = 20$.

Part (a): Determine $R(j\omega)$ in terms of $X(j\omega)$.

- **Step 1: Find $Y(j\omega)$.** Modulation shifts the spectrum:
  $Y(j\omega) = \frac{1}{2}[X(j(\omega - 30)) + X(j(\omega + 30))]$. This creates copies of $X(j\omega)$ centered at ± 30. The positive frequency part is non-zero for $10 < \omega < 50$.
- **Step 2: Find $W(j\omega)$.** Apply the band-pass filter:
  $W(j\omega) = Y(j\omega)H_{BP}(j\omega)$. The filter passes frequencies from $30 < |\omega| < 50$. This isolates the upper portion of the modulated spectrum. The resulting spectrum $W(j\omega)$ is what you would get by modulating $x_1(t) = \frac{\sin(10t)}{\pi t}$ (whose spectrum $X_1(j\omega)$ is a rect from -10 to 10) with $\cos(40t)$.
- **Step 3: Find $Z(j\omega)$.** Demodulation by $\cos(40t)$:
  $z(t) = w(t)\cos(40t)$ effectively demodulates the signal. $Z(j\omega) = \frac{1}{2}[W(j(\omega - 40)) + W(j(\omega + 40))]$. This process creates a baseband component and high-frequency components. The baseband component is $\frac{1}{2}X_1(j\omega)$, where $X_1(j\omega)$ is a rect from -10 to 10.
- **Step 4: Find $R(j\omega)$.** Apply the low-pass filter:
  The LPF has gain 2 and $\omega_c = 20$. It passes the baseband component of $Z(j\omega)$ entirely.
  $R(j\omega) = 2 \times (\frac{1}{2}X_1(j\omega)) = X_1(j\omega)$.
- **Step 5: Relate to $X(j\omega)$.** $X_1(j\omega)$ is a rect from -10 to 10. $X(j\omega)$ is a rect from -20 to 20. By frequency scaling, a signal with transform $X(j2\omega)$ would have its frequency axis compressed by a factor of 2. Thus, $X(j2\omega)$ is a rect from -10 to 10.
- **Final Answer (a):** $R(j\omega) = X(j2\omega)$.

Part (b): Determine the output signal $r(t)$.

- We found $R(j\omega) = X(j2\omega)$. We find $r(t)$ by taking the inverse transform. Using the time-scaling property $x(at) \leftrightarrow \frac{1}{|a|}X(j\frac{\omega}{a})$, we can see that the transform $X(j2\omega)$ corresponds to the time-domain signal $\frac{1}{2}x(t/2)$.
- $r(t) = \frac{1}{2}x(t/2) = \frac{1}{2}\frac{\sin(20(t/2))}{\pi(t/2)} = \frac{1}{2}\frac{\sin(10t)}{\pi t/2} = \frac{\sin(10t)}{\pi t}$.
- **Final Answer (b):** $r(t) = \frac{\sin(10t)}{\pi t}$.

2022 Problem 4 [Sampling Theory]

Given: Signal $x(t)$ is bandlimited to W, so $X(j\omega) = 0$ for $|\omega| > W$. Find the minimum sampling frequency ω_s to reconstruct $y(t)$.
Part (a): $y(t) = (x(3t - 1))^2$

- **Step 1: Bandwidth of $z(t) = x(3t-1)$.** The signal $x(t)$ is scaled in time by $a = 3$. This expands the bandwidth by a factor of 3. The time shift does not affect bandwidth. $x(3t) \leftrightarrow \frac{1}{3}X(j\frac{\omega}{3})$. Bandwidth becomes $3W$. $x(3t-1) = x(3(t-1/3)) \leftrightarrow \frac{1}{3}e^{-j\omega/3}X(j\frac{\omega}{3})$. Bandwidth is $W_z = 3W$.
- **Step 2: Bandwidth of $y(t) = z(t)^2$.** Squaring a signal in the time domain corresponds to convolving its spectrum with itself: $Y(j\omega) = \frac{1}{2\pi}Z(j\omega) * Z(j\omega)$.
- The bandwidth of the convolution of two signals is the sum of their individual bandwidths. $W_y = W_z + W_z = 3W + 3W = 6W$.
- **Step 3: Minimum Sampling Frequency.** The Nyquist rate requires the sampling frequency ω_s to be greater than twice the maximum frequency in the signal. $\omega_s > 2W_y = 2(6W) = 12W$.
- **Final Answer (a):** $\omega_s > 12W$.

Part (b): $y(t) = x(t) * \frac{\sin(2Wt)}{\pi t}$

- **Step 1: Identify the operation.** This is a convolution in the time domain, which corresponds to multiplication in the frequency domain: $Y(j\omega) = X(j\omega)Z(j\omega)$.
- **Step 2: Find the spectrum $Z(j\omega)$.** Let $z(t) = \frac{\sin(2Wt)}{\pi t}$. From the FT pair table, its transform is a rectangular pulse: $Z(j\omega) = \begin{cases} 1, & |\omega| < 2W \\ 0, & |\omega| > 2W \end{cases}$.
- **Step 3: Perform multiplication in frequency.** $Y(j\omega) = X(j\omega) \times Z(j\omega)$. We know $X(j\omega) = 0$ for $|\omega| > W$. In the range where $X(j\omega)$ can be non-zero (i.e., $|\omega| < W$), the value of $Z(j\omega)$ is always 1. Therefore, the product is simply $X(j\omega)$.
- **Step 4: Determine bandwidth of $Y(j\omega)$.** Since $Y(j\omega) = X(j\omega)$, the bandwidth of $y(t)$ is the same as the bandwidth of $x(t)$, which is W.
- **Step 5: Minimum Sampling Frequency.** $\omega_s > 2W_y = 2W$.
- **Final Answer (b):** $\omega_s > 2W$.

2021 Problem 6 [Block Diagram]

Causal LTI system defined by $\frac{d^2y(t)}{dt^2} - 2\frac{dy(t)}{dt} - 3y(t) = \frac{dx(t)}{dt} - x(t)$.

(c) (10 points)

Draw a block diagram for the direct form (canonical) realization of the system using a cascade of integrators with feedforward and feedback components.

$$H(s) = \frac{s-1}{s^2-2s-3} = (s-1)\frac{1}{s^2-2s-3}$$

$$H(s) = \underbrace{(s-1)}_{H_1(s)}\ \underbrace{\frac{1}{s^2-2s-3}}_{H_2(s)}$$

$Y(s) = H_1(s)F(s)$

$F(s) = H_2(s)X(s)$

$s^2F(s) - 2sF(s) - 3F(s) = X(s)$

$s^2F(s) = 2sF(s) + 3F(s) + X(s)$

$Y(s) = (s-1)F(s)$

$Y(s) = sF(s) - F(s)$